

Taiga Nishida

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EDUCATION

Carnegie Mellon University

3.9 GPA

Bachelor of Science in Computer Science, Concentration in Machine Learning

May 2027

Relevant Coursework:

Deep Reinforcement Learning, Generative AI, Compilers, Graduate Machine Learning, Deep Learning, Computer Systems, Data Structures and Algorithms, Functional Programming, Theoretical Computer Science, Multivariable Calculus, Honors Linear Algebra, Probability and Computation, Software Security and Privacy, Computer Graphics

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SML/NJ, OCaml, Java, JavaScript

Frameworks and Libraries: HuggingFace, Pandas, OpenCV, Numpy, TensorFlow, PyTorch, Matplotlib, Pillow, Scikit-Image, Scikit-Learn, Google Colab,

Technical Skills: Compute Cluster usage, COLMAP, TinyCUDANN, Unreal Engine, Git, Github, Linux, MeshLab, L^AT_EX

EXPERIENCE

AV Lab at CMU Robotics Institute under Dr. Deva Ramanan

May 2026 – Ongoing

- Designed and implemented a 3D reconstruction pipeline to perform **sparse novel view synthesis**, combining multi-resolution Hash-SDFs and background NeRFs for latent representation and combining ideas from MV-Sam3D, BA-NeuS, and Neuralangelo for the iterative optimization scheme.
- Fine-tuned the 600M-parameter MV-Sam3D flow transformer for specialization to buildings using **Low-Rank Adaptation (LoRA)**.

Hong Lab at Stanford CIBSR under Dr. David S. Hong

June 2023 – May 2024

- Proposed a direct **frequency-domain analysis** approach to replace a flawed pipeline combining **wav2vec2** and NLP-based text classification in a parent-child interaction therapy automation project.
- Designed the alternative as an improvement over **SpecialTime's OpenSMILE model**, the then state-of-the-art in automated behavioral speech analysis.

Dragonfruit AI

June 2022 – August 2022

- Helped overhaul the codebase, migrating from an outdated **YOLOv5** pose-detection model to the then-state-of-the-art **MoveNet** architecture.
- Proposed and prototyped adoption of **BlazePose** as a potential next-generation improvement beyond MoveNet.

PROJECTS

City Procedural Modeling | *Automatic city model generation from satellite imagery* Spring 2021 – Spring 2023

- Used the **U-Net** model to **segment satellite images** into classes such as buildings, trees, cars, etc.
- Applied **procedural modeling** to automate the construction of 3D city replicas from segmentation outputs, generated using **Unreal Engine**.

Facade Window Detection | *Recursive deep learning for building analysis*

Spring 2019 – Spring 2020

- Proposed and implemented a novel recursive approach for detecting window alignments in building facades using a **VGG19**-based model with iterative cropping to refine detections along horizontal and vertical alignments.

Sidewalk Detection | *Satellite-based pedestrian path detection*

Spring 2018 – Spring 2019

- Trained a U-Net model to detect sidewalks from Google satellite imagery to improve Google walking routes.

HONORS

Platinum Competitor in USA Computing Olympiad (USACO)

2021 – 2024

Four Time USA(J) Mathematical Olympiad (USA(J)MO) Qualifier via AIME

2020 – 2023

Qualified for Broadcom MASTERS Fair with Window Detection Project

2020

1st Place in USA Physics Research Tournament

2021